Chapter 0 Part B – Problem Solving in Physics

Calculus Physics II

General Steps for Problem Solving in Physics

Below is an outline for solving **problems in Physics**, it is only a framework to approach problems, each problem will be different and require its own detailed approach. Problem solving in physics involves a systematic and analytical approach that blends conceptual understanding with mathematical reasoning. Throughout the process, critical thinking and logical reasoning are essential, as is the ability to evaluate results for consistency and physical plausibility. Ultimately, the goal is not just to find a solution, but to develop a process and deepen understanding of the underlying physical phenomena.

By the end of this course, students will be able to:

I. Organize the information presented in problems about various mechanical phenomena into sketches, diagrams, and/or tables representing the problem situation and the physical quantities involved.
II. Identify the fundamental principles of physics underlying these phenomena.
III. Formulate mathematical equations using only symbols to describe the relationship between physical quantities in these problems.
IV. Solve these mathematical equations to find the values of the physical quantities and/or to predict system behavior.
V. Evaluate whether answers are reasonable given the constraints of the particular problems being solved.

Problem solving in Physics

1. Understand the Problem
   - Read the problem carefully and make mental notes of the object of interest and what is happening to it.
   - Visualize the problem, image what the object is doing and why.
   - Determine the conceptual tools needed to solve the problem.
   - Draw a diagram or sketch and label the types of motion, forces, initial and final positions, etc. …
   - Identify and define the knowns and unknowns (with proper units) given in the problem also implied values like “starts from rest”.
   - Assign variables with mnemonic symbols, symbols and subscripts that identify the variables in the problems, so you do not get variables mixed up.

2. Determine the relevant Physics for solving the Problem.
   - What physics does the problem involve, type of motion, Forces, Energy, etc. ...
   - Identify vectors and scalar quantities.
   - Determine if any of the quantities have a dependance on time, displacement or other quantities, and are the quantities average or is calculus going to be needed.
   - Start with general Physics concepts and then narrow down using the information in the problem.
   - Break the problem into parts or multiple steps if needed.
   - Determine the type of math needed to solve the problem (algebra, derivatives or integrals). Think about the steps needed to isolate the unknown from the given information.

3. Plan the solution
   - Set up appropriate coordinate System that simplifies the math and makes sense for the problem.
   - Label diagram in the chosen coordinate system with each variable including the magnitudes and directions.

- Consider if there may be other approaches and which appears to make the most sense to use for this problem. There may be more than one path that is correct.
- Outline the path to a solution using words in terms of the physics involved.
- Do NOT equation hunt or page through your notes to find the "magic" formula!

4. Execute Plan
   - Write out the equations in the most general terms to solve the problem.
   - Write your solutions logically and neat on the page, write each step with an explanation. Do not do the math in your head but write it out as you go to minimize mistakes.
   - Solve equation using symbols only for the unknown (this may take multiple steps and calculus!).
   - Do NOT plug in numbers until a final equation has been derived!
   - Re-write the equation eliminating any variables that have a value of zero or trig functions that will have a value of 1, etc. …
   - Simplify the equation and use the known variables in the solution.

5. Evaluate your answer.
   - Verify that the equation gives the correct units for the given unknown.
   - Does the equation give the correct results using different conductions or known values or extremes?
   - If numbers are used, does the value make sense physically?
   - If any of the answers to the questions above is NO, then go back to step 2.

6. State the solution and reflect on results.
   - Box or highlight final answer.
   - Include proper units, significant figures and direction if applicable.
   - What principles were important in solving the problem.

Note: The problems that are solved in class, in the lecture notes and in problems solutions are very valuable in your learning and should be studied but they are the end product of the instructor's problem solving and do not show the true process involved in reaching the end product. Students should expect to struggle with problems solving in physics.

Watching an instructor's problem-solving videos for this Physics class is highly beneficial because it transforms abstract theoretical concepts into tangible, procedural knowledge, bridging the gap between textbook formulas and actual application. These videos allow students to observe an expert's thought process—such as how to break down complex mechanics problems, draw free-body diagrams, and choose the correct physical principles (like conservation of energy or momentum)—rather than just focusing on "plug and chug" math. By offering on-demand, self-paced learning, students can pause and review difficult sections, reducing cognitive load and lowering anxiety, which is particularly useful for building confidence before high-stakes exams. Furthermore, seeing the instructor work through problems step-by-step helps students identify common mistakes and adopt structured, effective strategies for tackling similar homework or exam questions.

Benefits of Watching Problem-Solving Videos in Physics. Posted on Canvas are general problem solving techniques for each of the modules, they are there too:

1. Reinforcement of Conceptual Understanding - Problem-solving videos emphasize the reasoning behind each step, not just the final answer. Connect mathematical procedures to underlying physical principles (e.g., how Newton's Second Law guides equation setup).

2. Videos provide access to expert problem-solving strategies that are often implicit in textbook solutions. This supports the development of metacognitive skills, learning how to think like an expert problem solver.

3. Step-by-Step Problem Decomposition, unlike static solutions, videos show the dynamic process of solving problems breaking complex problems into manageable steps and revising the approach when initial strategies fail.

4. Self-Paced and Repeatable Learning Students can pause, rewind, and rewatch difficult segments. This flexibility supports diverse learning speeds and promotes mastery learning.

5. Can get immediate Feedback and Error Correction, when attempting a problem first and then watch the solution they can identify and correct misconceptions. They gain insight into common mistakes and how to avoid them and learn how small errors propagate through a solution.

6. The videos integrate the visual and analytical representations of the problem using diagrams, graphs, and equation with calculus and algebraic steps.

7. The videos make problem solving feel more approachable, increased Engagement and Motivation.

8. Bridging Gaps in Mathematical Preparation - Students with weaker calculus or algebra backgrounds can see how mathematical tools are applied in context and can build confidence in manipulating equations within physical problems.

Best Practices for watching Problem solving videos - To maximize effectiveness, students should not passively watch videos. Instead, they should:

1. Attempt the problem first.
2. Watch actively (pause and predict the next steps).
3. Take notes on reasoning, not just steps.
4. Reflect on differences between their approach and the video.

Student Problem-Solving Checklist: General Approach

Use this checklist for every problem to ensure a clear, structured, and correct solution.

- ☐ Read the problem and identify the physical principle(s) involved (e.g., Newton's Laws, Energy, Momentum).
- ☐ Define the system and draw a clear, labeled diagram and/or a force diagram.
- ☐ Choose an appropriate coordinate system, define directions and label.
- ☐ List all known (given) quantities and unknown variables.
- ☐ Translate the situation into the general mathematical equations that may be used.
- ☐ Solve symbolically before substituting numerical values.
- ☐ Substitute values carefully with correct units.
- ☐ Check units and dimensional consistency throughout the solution.
- ☐ Evaluate whether the answer is physically reasonable (magnitude and direction).
- ☐ Reflect: What concept was key? How would the solution change under different conditions?

Instructor Note: Full solutions should include all steps above for full credit

Problem solving Rubric given on next page!

Rubric for grading problems solved in Physics 2 – Note the "Criteria" that the grade is based on and amount of point assigned to each Criteria. Ony "Mastery" and "Competent" are passing grades. This Rubric will be applied to the following assessments: Problem solving on Exams, and Worksheet Part B – worked out problems.

Criteria	Mastery	Competent	Minimum Understanding	Unacceptable
Useful Description	The description and information from the problem is appropriate and complete	The description and information from the problem contain mirror errors or omission	The description and information from the problem contains significant errors or omissions	There is no description or information from the problem given.
Points	10	8	4	0
Physics Approach	Selected the relevant physics concepts and principles.	Most of the relevant physics concepts and principles were selected with some minor omissions and errors.	Major errors and omissions when selecting the relevant physics concepts and principles.	No concepts and principles were indicated.
Points	30	20	10	0
Specific Applications of Physics	The appropriate application of the Physics was applied to the specific conditions in this problem and the general equations used	Most of the appropriate application of the Physics were applied to the specific conditions in this problem expect for some minor omissions and errors.	Major errors and omissions when applying the Physics to the specific conditions in this problem.	The solution does not indicate an application of the physics.
Points	30	20	10	0
Mathematical Procedures	Applied the correct mathematical rules and procedures in solving the problem.	Applied the correct mathematical rules and procedures in solving the problem except for some mirror errors or omissions.	Major errors and omissions when applying the mathematical rules and procedures in problem solving.	Did not apply any of the mathematical rules or procedures in problem solving.
Points	20	15	5	0
Logical Organization	The entire problem is clear, focused and logically connected.	The solution is clear and focused with mirror inconsistencies.	The majority of the solutions is unclear with little focus and connection.	There is no evidence of logical progression and no focus or connection.
Points	10	8	4	0

Based on “Assessing student written problem solutions: A problem-solving rubric with application to introductory physics” Jennifer L. Docktor, Jay Dornfeld, Evan Frodermann, Kenneth Heller, Leonardo Hsu, Koblar Alan Jackson, Andrew Mason, Qing X. Ryan, and Jie Yang Phys. Rev. Phys. Educ. Res. **12**, 010130 – Published 11 May 2016.